LIFE SCIENCE · PART A · TRIMESTER 1

Cells, Microscopes and the Lab: Review

ANSWER KEY

Answer in complete sentences when it asks you to explain. Answers are in the answer key that comes with this worksheet in Classboard.

A · LAB SAFETY AND TOOLS

1. Your partner spills an unknown liquid on the table. List two things you should do.

Tell the teacher right away; don't touch it or clean it up with your hands; move away from it; follow the teacher's directions.

2. Which tool would you use to measure exactly 25 mL of water? How do you read it correctly?

A graduated cylinder. Put your eyes level with the water and read the bottom of the meniscus.

B · MICROSCOPES

3. Fill in the total magnification for a 10× eyepiece:

Objective	Total magnification
4×	
10×	
40×	

40× · 100× · 400×

4. Why should you only use the fine focus knob on high power?

The coarse knob moves the stage too far and can crash the objective into the slide (and lose the focus).

5. Put these steps in order: __ switch to a higher objective · __ place the slide on the stage · __ focus with coarse, then fine · __ start on the lowest objective · __ sharpen with fine focus only

4 · 1 · 3 · 2 · 5 (place slide → lowest objective → coarse then fine → higher objective → fine only)

C · SCIENCE SKILLS

6. Leo tests if music helps bean plants grow. Group A hears music; group B doesn't. Name the independent variable, dependent variable, control group and one controlled variable.

Independent: music (or not). Dependent: how much the plants grow. Control group: group B (no music). Controlled: any one of water, light, soil, plant type, pot size.

D · LIVING THINGS AND CELL THEORY

7. Write the three parts of the cell theory.

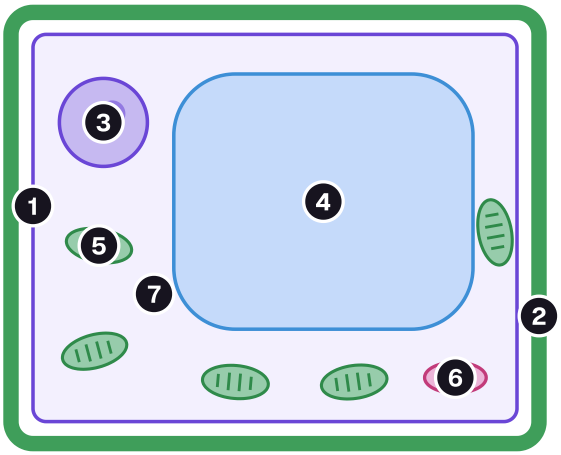
   1. All living things are made of one or more cells. 2. The cell is the basic unit of structure and function. 3. All cells come from existing cells.

8. Claim, evidence, reasoning: Is a crystal that "grows" in salt water alive?

   Claim: No. Evidence: a crystal is not made of cells, doesn't use energy, respond or reproduce. Reasoning: living things must have all the characteristics of life, and growing alone isn't enough.

E · CELL PARTS

9. Name each numbered part of this plant cell.



   1 Cell membrane · 2 Cell wall · 3 Nucleus · 4 Central vacuole · 5 Chloroplast · 6 Mitochondrion · 7 Cytoplasm

10. Match each part to its job: nucleus, mitochondria, chloroplast, cell membrane, ribosome.
    a) makes proteins · b) control center · c) controls what enters and leaves · d) releases energy from food · e) makes food from sunlight

    Nucleus b · Mitochondria d · Chloroplast e · Cell membrane c · Ribosome a

11. Name three parts found in plant cells but not animal cells, and two parts found in both.

    Plant only: cell wall, chloroplasts, large central vacuole. Both (any two): cell membrane, nucleus, cytoplasm, mitochondria, ribosomes.

F · LEVELS AND BODY SYSTEMS

12. Put in order from smallest to largest: organ system, cell, organism, organ, tissue.

    Cell → tissue → organ → organ system → organism

13. Give an example of each level for the human body: tissue, organ, organ system.

    Example: heart muscle tissue → heart → circulatory system (other correct examples are fine)

14. You start running. Explain how your respiratory and circulatory systems work together.

You breathe faster to take in more oxygen (respiratory), and your heart beats faster to carry that oxygen in the blood to your working muscles (circulatory). Both also remove carbon dioxide.